DVD and can be download from http://tr.im/dtraceguide.

5.1 Overview

DTrace is a comprehensive dynamic tracing facility that is built into the Solaris OS and can be used by administrators and developers to examine the behavior of both user programs and of the operating system itself. With DTrace you can explore your system to understand how it works, track down performance problems across many layers of software, or locate the cause of aberrant behavior. DTrace is safe to use on production systems and does not require restarting either the system or applications.

DTrace dynamically modifies the operating system kernel and user processes to record data at locations of interest, called probes. A probe is a location or activity to which DTrace can bind a request to perform a set of actions, such as record a stack trace, a timestamp, or the argument to a function. Probes are like programmable sensors in your Solaris system. DTrace probes come from a set of kernel modules called providers, each of which performs a particular kind of instrumentation to create probes.

You can access DTrace capabilities by using the `dtrace` command or by using DTrace scripts. You must be the super user on the system to execute the `dtrace` command or DTrace scripts, or you must have appropriate DTrace privileges. DTrace includes a new scripting language called D that is designed specifically for dynamic tracing. With D, you can easily write scripts that dynamically turn on probes and collect and process the information. By sharing D scripts, you can easily share your knowledge and troubleshooting methods with others.

5.2 Introduction to D Scripts

DTrace works on an event/callback model: You register for an event and implement a callback that is executed when the event occurs. Examples of events include executing a particular function, accessing a particular file, executing an SQL statement, garbage collecting in JavaTM. Examples of data collected in the callback include function arguments, time taken to execute a function, SQL statements. In DTrace the event definition is called a probe, the occurrence of the event is called probe firing, and the callback is called an action. You can use a predicate to limit the probes that fire. Predicates are statements that evaluate to a boolean value. The action executes only when the predicate evaluates to true. A D script consists of a probe description, a predicate, and actions as shown below: